

Bitcoin's Price Power Law: Stability, Maturation, and a Live Test of the Scaling Hypothesis

An empirical study of exponent stability, residual persistence, structural breaks, volatility decline, and below-trend recovery

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Data through August 14, 2026

Abstract

Bitcoin's historical price has long been noted to exhibit an unusually close power-law relationship with time. Recent work has formalized this observation, examined the stability of the fitted exponent, and proposed mechanisms linking price scaling to network adoption. This paper does not propose a new Bitcoin Power Law and does not attempt to validate a particular mechanistic derivation. Instead, we reproduce the core empirical relationship and use it as the starting point for a narrower set of stress tests motivated by Bitcoin's maturation and its 2025-2026 below-trend episode.

Using 5,873 completed UTC daily closes from July 17, 2010 through August 14, 2026, and defining time as days since Bitcoin's January 3, 2009 genesis, we estimate

$$\ln P(t) = \ln A + B \ln t$$

with $A = 4.156832 \times 10^{-17}$, $B = 5.650938$, and $R^2 = 0.961281$. The fit is descriptively close in log-log space, but its residuals are extremely persistent, with lag-1 autocorrelation of 0.996985. Accordingly, ordinary least squares (OLS) uncertainty is misleading: a Newey-West estimate using a 365-day lag gives a standard error for B of approximately 0.189, versus 0.0148 under naive OLS, while a 365-day moving-block bootstrap produces a standard deviation of approximately 0.240. This independently corroborates the similarly widened autocorrelation-aware uncertainty reported by wineLightning and Claude AI [9]. The numerical exponent is also conditional on the genesis-date time origin. Baquero and Menezes [2] report that plausible additive shifts of the time coordinate can change the fitted exponent by nearly a factor of three, so B should be read as a within-convention estimate rather than a shift-invariant structural constant.

A Bayesian Information Criterion (BIC) segmentation procedure is uninformative about a modern structural break once the detector is calibrated against a constant-exponent null with persistent residuals. In 300 Monte Carlo null simulations, the detector selects the maximum permitted four breaks in 99.7% of cases. More importantly, the size of the real-data breaks is ordinary under the same null: the largest adjacent slope change in the real series sits at only the 54.7th percentile of the simulated distribution. In plain terms, the break detector behaves much the same in a world where the exponent is held constant.

The current below-trend episode presents a more nuanced result. Its maximum raw log residual, -0.8165, is almost identical to the 2022 minimum and is historically mid-field by raw depth. Yet residual volatility is substantially lower in the modern market. The same raw gap is therefore large relative to today's smaller typical residual fluctuations. At the June 30, 2026 episode low, raw distance from the trend divided by trailing four-year residual standard deviation reached a local-sigma trend depth of -2.732; by the August 14 cutoff it had improved to -2.521. The episode ranks deepest among seven modern episodes with sufficient trailing history under that measure. A conventional locally de-measured z-score also ranks the episode first by minimum, but only narrowly: -1.907 versus

-1.847 in 2022. The distinction arises because the recent residual distribution itself is persistently centered below zero.

We further test the hypothesis that declining volatility causes slower recovery toward the Power-Law trend. The hypothesis is not supported. Lower variance mechanically makes a fixed raw gap more extreme in standardized units, but under fixed-persistence AR(1) dynamics it cannot slow first passage from an equally standardized gap. Rolling estimates of residual persistence do not display a stable monotonic relationship with declining volatility, and historical recovery episodes are too few to identify a reliable volatility effect.

Finally, we distinguish the return required merely to keep pace with the trend from the substantially larger return required to erase an accumulated deficit. The current one-year trend hurdle is approximately 36.6%, near the 35th percentile of Bitcoin's historical rolling one-year CAGR distribution. Closing the existing gap requires approximately 199% over one year, 100.5% annualized over two years, or 74.6% annualized over three years. The current challenge is therefore less the trend's growth rate than the accumulated distance below it.

Taken together, the results do not establish that Bitcoin's historical Power Law is structurally immutable, but neither do the completed tests provide compelling evidence of structural failure. The current episode remains a meaningful live test of the empirical scaling relationship.

1. Introduction

Bitcoin's price history is unusual even by the standards of financial time series. Its growth spans several orders of magnitude, its return distribution has changed materially as the asset has matured, and its market structure has evolved from a thin retail market into one that includes regulated derivatives, exchange-traded products, institutional custody, and sovereign and corporate ownership.

Against that changing background, a comparatively simple empirical observation has persisted: plotting Bitcoin's price against its age on logarithmic axes produces an unusually linear relationship.

A common representation is

$$P(t) = A t^B,$$

or equivalently,

$$\ln P(t) = \ln A + B \ln t,$$

where $P(t)$ is Bitcoin's price, t is age measured from a specified origin, A is an intercept parameter, and B is the fitted scaling exponent.

The empirical relationship has received increasing attention. Santostasi and Perrenod [1] report a fitted exponent near 5.69 and develop a mechanistic interpretation based on network adoption and generalized Metcalfe scaling. Other researchers have examined residual stationarity, exponent stability, scale invariance, and alternative structural interpretations [2,3].

Our investigation began for a less ambitious reason.

While updating a Bitcoin Power Law dashboard, we observed that the fitted value of B had moved. Because B directly affects forward extrapolations, a simple engineering question became an empirical one:

Why is B changing?

That question led to several others.

Is the apparent convergence of B evidence of a stable structural exponent, or would a progressively more stable estimate arise mechanically as the sample grows? Do statistical break detectors identify genuine slope changes, or merely Bitcoin's highly persistent cycle-like residual structure? Is the current below-trend episode historically unusual? Does its severity change when modern volatility is taken into account? Is the fitted model "rescuing" itself by moving downward as price underperforms? Has Bitcoin's declining volatility made recovery to trend mechanically more difficult? And, perhaps most importantly, how different is the return required to **keep pace** with a rising Power-Law trend from the return required to **catch up** after a large accumulated deficit?

Several plausible explanations did not survive testing.

That is an important feature of the study rather than an inconvenience. The objective was not to demonstrate that the Power Law must hold. It was to expose the relationship to questions that could plausibly weaken it.

The resulting contribution is therefore narrower than proposing a new model. We take the existing empirical Bitcoin Power Law literature as a starting point and extend it with a set of focused diagnostics around:

1. independent corroboration of robust uncertainty in the expanding exponent estimate;
2. structural-break detection under a matched persistent-residual null;
3. systematic comparison of below-trend episodes;
4. the distinction between raw trend distance and locally standardized trend distance;
5. decomposition of current underperformance into model-refit absorption and remaining price shortfall;
6. the relationship between volatility, persistence, and recovery speed;
7. the declining growth requirement of the trend; and
8. the difference between keeping pace with the trend and closing an existing deficit.

The findings are mixed in a useful way. The historical fit remains descriptively strong. The current episode does not look unprecedented in raw distance. The recent decline in B is real numerically but not large relative to robust uncertainty. Structural-break tests lose their apparent significance once persistence is incorporated into the null. Declining volatility does not provide a convincing explanation for slower recovery. At the same time, the modern residual distribution has shifted materially below the long-run trend, and recent multi-year realized growth has at times lagged the contemporaneous Power-Law hurdle.

The appropriate conclusion is therefore neither "the Power Law is proven" nor "the Power Law has failed." It is that the completed stress tests do not provide compelling evidence of structural failure, while the current episode continues to test the relationship.

2. Related Work and Positioning

2.1 The empirical Bitcoin Power Law

Santostasi and Perrenod [1] provide the recent formal account of an empirical observation that Santostasi traces to earlier informal work beginning in the prior decade. They report a regression approximately equivalent to

$$\log_{10} P(t) = -16.509 + 5.690 \log_{10} t,$$

with $R^2 \approx 0.961$, using data through February 18, 2026.

Using our independent daily series and stopping on the same date, we obtain

$$B \approx 5.6912,$$

which is extremely close to their reported 5.690.

This reproduction is useful for a simple reason: it establishes that the empirical object examined in this paper is essentially the same price-age relationship examined in their work rather than an artifact of our dashboard implementation or data pipeline.

It does **not** validate their mechanistic derivation.

Santostasi and Perrenod relate the price exponent to network-adoption and generalized Metcalfe exponents. We do not independently estimate those network components. Our work therefore concerns the stability and behavior of the **empirical price-age scaling relation**, not proof of the mechanism proposed to explain it.

2.2 Stability, residuals, and competing interpretations

Recent literature has already moved well beyond simply fitting a straight line on logarithmic axes.

Santostasi and Perrenod examine local slope stability and sequential convergence. Perrenod [3] separately studies residual persistence, stationarity, and declining residual amplitude: whether deviations from the trend tend to linger, whether those deviations behave as though they remain anchored to a stable long-run process, and whether their typical magnitude has shrunk over time.

Baquero and Menezes [2] provide an especially important skeptical counterpoint. Their analysis questions strong structural interpretations in three ways that matter here. First, the fitted time-domain exponent is highly sensitive to additive shifts in the assumed time origin. Second, residual and scale-invariance diagnostics used in earlier work as evidence supporting the Power Law also pass on a fitted multi-component sigmoid alternative, so those diagnostics do not uniquely identify a structural Power Law. Third, their quarterly wave-stability bootstrap rejects a Power-Law-plus-AR(1) null for Bitcoin price at $p=0.015$, although the result is not Bonferroni-robust.

At the same time, their walk-forward analysis finds that the simple Power Law performs unusually well at 12-24 month horizons relative to flexible trend models and standard time-series baselines. That combination is important. Their result is not "the Power Law is structurally proven" or "the Power Law is useless." It is closer to: the model can remain useful as a long-horizon empirical description even when uniqueness and structural mechanism are not established. A close descriptive fit and useful forecasting properties do not by themselves imply a unique structural law.

Our paper therefore avoids treating R^2 as inferential evidence. With residual lag-1 autocorrelation near 0.997, high R^2 is not sufficient to establish a structural relationship.

2.3 Volatility, envelopes, and recovery

Declining Bitcoin volatility is widely documented. Perrenod [3] also reports declining amplitude specifically in deviations around the Power-Law trend. Practitioner implementations have proposed moving or empirical bands rather than treating fixed standard-deviation bands as Gaussian probability statements.

Those moving-band approaches are relevant motivation, but they are not the hypothesis tested here. We do **not** test whether fixed bands or moving bands are preferable. Instead, we test whether the decline in residual volatility itself changes the mechanics of recovery toward the trend.

Banks [4] examines separate upper and lower Power-Law envelopes and reports materially different scaling behavior at the upper boundary, while other work on cryptocurrency return tails finds maturation-related changes and asymmetries in extreme returns [5].

A separate line of analysis has modeled deviations from a Power-Law trend as mean-reverting processes. One 2026 practitioner study uses a rolling Ornstein-Uhlenbeck framework and argues that mean reversion may slow as Bitcoin matures [6]. We treat this as a hypothesis to be tested rather than an established result. Mean reversion here is a statistical tendency, not an assumption that price must return to trend.

A different line of work treats cyclical deviations as containing deterministic log-periodic structure rather than generic persistent noise. Wheatley et al. [7] combine a generalized-Metcalf value model with a Log-Periodic Power Law Singularity specification for Bitcoin bubbles, while Perrenod [8] later applies log-periodic modes directly to residuals around a Bitcoin price Power Law and reports substantial residual-variance reduction. We do not test a log-periodic residual specification here. Our break-null and AR(1) mechanism results should therefore be read as conditional on the persistent-stochastic residual models actually examined.

2.4 Where this study extends prior work

We do not claim novelty for exponent stability, residual persistence, declining volatility, or the general idea of recovery toward trend. Those questions already exist in the literature.

A July 2026 study by wineLightning and Claude AI [9] independently reports materially wider autocorrelation-aware uncertainty for the time-domain exponent and also examines time-origin sensitivity. We therefore treat our robust-uncertainty result as independent corroboration rather than a novelty claim.

In our search, however, we did not identify prior work that explicitly combines the narrower analyses used here in this form, particularly:

- ranking complete below-trend episodes jointly by duration, depth, cumulative area, and modern volatility-scaled trend depth;
- quantifying how much of a live trend deficit is absorbed by regression refitting;
- explicitly separating **distance from the model's trend in units of recent volatility** from a conventional locally de-measured z-score;
- testing the volatility-recovery hypothesis separately through scale, standardized depth, and persistence channels; and
- distinguishing the growth required to **keep pace** with the current trend from the growth required to **catch up** to it, and benchmarking those quantities against historical rolling returns.

The third distinction is especially important. A trend-centered measure asks, "How far below the model are we, relative to the size of modern residual fluctuations?" A locally de-measured z-score asks, "How unusual is this observation relative to where residuals have recently been centered?" These are related but different questions.

We therefore frame the paper as an extension and stress test of prior work rather than a new Power-Law model.

3. Data

3.1 Price series

The canonical series contains completed UTC daily Bitcoin closes from:

- July 17, 2010 through August 14, 2026;
- $n=5,873$ observations;
- zero missing dates; and
- zero duplicate dates.

The current partial UTC day and all intraday spot observations are excluded from model fitting.

This distinction matters because the original dashboard had previously allowed current-day information to enter paths where only completed historical observations should have been used. The production data pipeline was repaired before any research reported here was conducted.

All completed studies subsequently used a byte-identical historical series.

3.2 Time variable

We define

$$t_i = \text{completed days since January 3, 2009.}$$

This is the Bitcoin-genesis convention used by the model under study.

The choice is substantively important. A power law in t is not invariant to additive shifts of the time coordinate. Baquero and Menezes [2] report that the fitted exponent changes by nearly a factor of three across their shift-parameter sweep, even though the genesis-date convention is favored by their information-criterion comparison.

We therefore set t_0 to January 3, 2009 as a **model convention**, not as an innocuous normalization. All reported values of B are conditional on this choice.

3.3 Returns and residuals

The fitted residual is

$$r_t = \ln P_t - (\ln A + B \ln t).$$

Daily log returns are

$$g_t = \ln P_t - \ln P_{t-1}.$$

Residual volatility, return volatility, persistence, episode depth, and local-standardization measures are derived from these quantities.

4. Baseline Model

The full-history regression is

$$\ln P_t = \alpha + B \ln t + \varepsilon_t,$$

where $\alpha = \ln A$.

The full-sample estimates are:

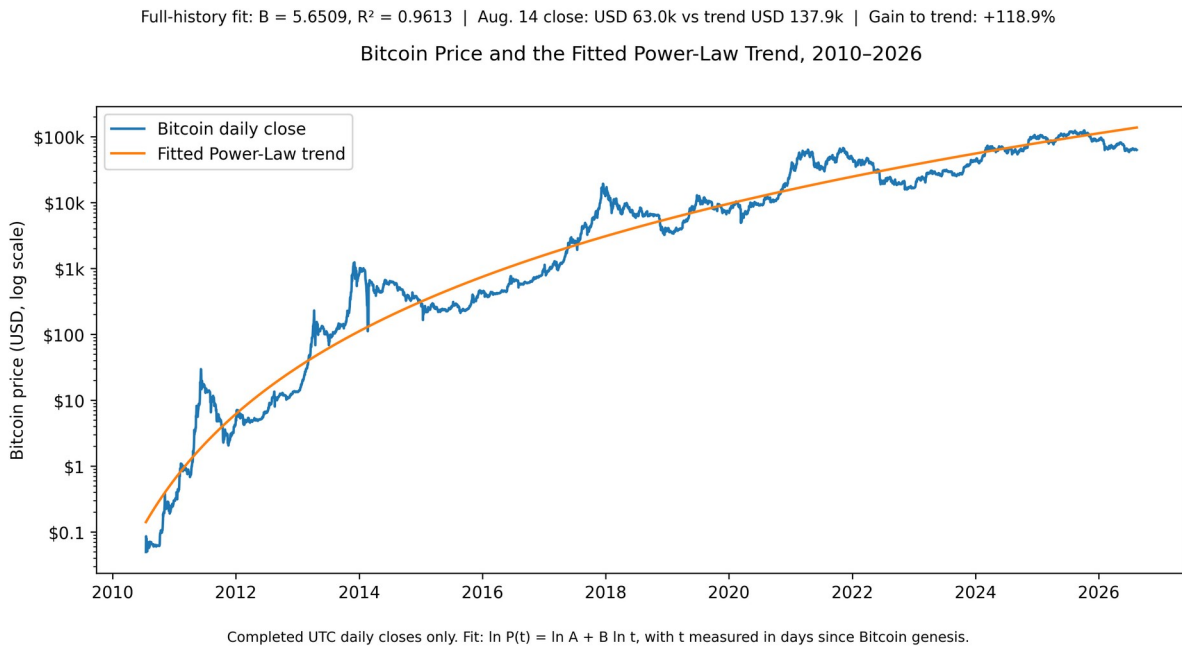
Quantity	Estimate
A	4.156832×10^{-17}
B	5.650938
R^2	0.961281

Quantity	Estimate
residual SD	0.697088
residual lag-1 autocorrelation	0.996985

The fit is descriptively close. That statement should not be confused with a claim of conventional statistical significance or with small price-space errors.

"Descriptively close" means that the fitted log-log line tracks the historical data closely across roughly six orders of magnitude of price. A residual standard deviation of 0.697 in natural-log units corresponds to a one-standard-deviation multiplicative scale of $e^{0.697} \approx 2.01$ in price. Baquero and Menezes [2], using a nearby CoinDesk window, report residual dispersion of about 0.30 in $\log_{10}$ units, or about 0.69 in natural-log units, which is closely comparable. Statistical inference is a different question again. The residual autocorrelation is so large that the standard independent-and-identically-distributed assumptions underlying naive OLS uncertainty are badly violated. In particular, the usual tiny OLS standard errors should not be read as evidence that the exponent is known with comparable precision.

Figure 1. Bitcoin Price and the Fitted Power-Law Trend, 2010-2026



5. Methods

5.1 Sampling-frequency robustness

The regression is re-estimated using daily, weekly, and monthly observations. The resulting estimates of B differ by less than approximately 0.02.

This check asks whether the headline exponent is an artifact of dense daily sampling.

5.2 Expanding-window exponent estimates

We estimate B repeatedly using expanding samples ending at successive dates.

Because residual persistence makes naive standard errors unreliable, uncertainty is evaluated using:

1. Newey-West **heteroskedasticity-and-autocorrelation-consistent (HAC)** covariance estimates across a range of lag choices; and
2. a moving-block bootstrap of the $(\ln t, \ln P)$ series using several block lengths.

After this first definition, we use **HAC** as shorthand for heteroskedasticity-and-autocorrelation-consistent uncertainty estimates.

The 365-day HAC estimate is used as a practical headline value because the estimated uncertainty reaches a broad plateau around that scale.

5.3 Residual stationarity diagnostics

We report:

- Augmented Dickey-Fuller (ADF) tests across several lags;
- a Phillips-Perron statistic; and
- an Engle-Granger-style residual test using residual-based critical values.

In plain language, these tests ask whether deviations from the fitted trend behave more like a process that remains anchored over time or like a process that can wander persistently without returning to a stable center. The ADF test is one standard way to test for this unit-root-like behavior, with the result depending in part on how much lagged information is included.

These tests should be interpreted cautiously. $\ln t$ is a deterministic function of time, not a stochastic $I(1)$ series, so conventional cointegration terminology is not a perfect match to this setting. The tests are best read as diagnostics of persistence and unit-root-like behavior around a deterministic log-log trend.

5.4 Structural-break testing

Monthly observations are segmented using exact dynamic-programming piecewise regressions with:

- minimum segment length of 12 months;
- number of breaks K selected by BIC; and
- $K \in \{0, 1, 2, 3, 4\}$.

That last line simply means that the detector is allowed to choose anywhere from **zero through four breaks**, with BIC selecting the preferred level of complexity. The Bayesian Information Criterion balances fit against complexity: additional segments will almost always improve in-sample fit, so BIC makes each added break pay a penalty before it is retained.

The cap of four is pragmatic rather than theoretical. Four breaks already allow the algorithm to fit five separate regimes while keeping each segment long enough to be interpretable. We did not test higher caps in this study. This matters because the real-data detector selects the cap itself, so the result should not be interpreted as evidence that "four" is the true or optimal number of breaks.

The important control is not the detector's output on real data alone.

We generate a constant-true- B null and run the identical detector on each simulated series. The monthly null is calibrated to an AR(1) residual process with $\rho=0.9160$ and residual standard deviation 0.7001, while the true monthly slope is held fixed at $B=5.6346$. These are Monte Carlo null simulations: repeated synthetic histories in which the true slope is constant but the residuals retain the persistence and scale of the observed monthly residual series.

We compare:

- selected break count;
- maximum adjacent segment-slope change; and
- total range of fitted segment slopes.

The central question is therefore a control-experiment question: **would the same break detector still report apparently important breaks in a world where the exponent never actually changes?**

5.5 Below-trend episode definition

A below-trend episode is constructed from strict crossings of the zero-residual line. The canonical episode inventory requires at least 60 days below trend; 3-day and 7-day bridge tolerances are used as robustness checks for short interruptions around zero. Under the strict rule, the sample contains ten major below-trend episodes: nine completed episodes and the current open episode. Seven of the ten have at least four years of prior residual history and can therefore be scored on the local-standardization measures in Section 5.6.

For the historical taxonomy, residuals are measured against the current full-history fitted A and B as an ex-post common yardstick. This fixed-model construction is deliberately different from the separate no-lookahead diagnostic used to ask when the current episode would have been recognized in real time.

Episodes are summarized using:

- duration;
- average residual depth;
- maximum raw residual depth;
- cumulative negative residual area; and
- modern local-standardized depth.

The current episode is included in historical rankings but is distinguished from completed episodes where recovery-time analysis is required, because the current episode has not yet completed a recovery.

5.6 Three standardized trend-depth measures

Declining residual volatility creates an interpretive problem. A raw residual of -0.8 may represent a common fluctuation in one era and an unusually large move in another.

All trailing local statistics use observations strictly prior to day t , so the denominator and local mean do not incorporate the day being scored.

We therefore distinguish three measures.

Trend-centered local-sigma depth

$$D_{\sigma,t} = \frac{r_t}{s_{4y,t}},$$

where $S_{4y,t}$ is the trailing four-year residual standard deviation.

In plain language, this asks: **How many modern-sized residual fluctuations below the Power-Law trend is today's price?** A value near -2 means the price sits about two recent residual standard deviations below the model's zero line.

This measures distance from the Power-Law trend's own zero line in units of recent residual fluctuation. It is **not** a z-score because it does not subtract the recent residual mean.

Locally de-measured z-score

$$z_t = \frac{r_t - \mu_{4y,t}}{S_{4y,t}}.$$

where $\mu_{4y,t}$ is the trailing four-year residual mean. This asks a different question: **How unusual is today's residual compared with the recent regime itself?** If the entire recent residual distribution has shifted below the long-run trend, the de-measured z-score treats that shifted center as the local baseline.

RMS trend depth

$$D_{RMS,t} = \frac{r_t}{\sqrt{E_{4y}(r^2)}}.$$

This retains zero as the model reference point while increasing the denominator when the recent residual distribution persistently shifts away from zero. It is therefore a middle ground between the first two measures.

The distinction is important because the recent four-year residual mean is substantially negative. The trend-centered measure asks how far Bitcoin is from the model; the de-measured measure asks how far it is from recent experience.

5.7 Frozen-fit versus refit decomposition

To test whether the Power Law is merely "chasing" underperforming price, we freeze A and B at the beginning of the current below-trend episode and compare the resulting trend value with the trend produced by the current refit.

This analysis is intentionally scoped to the **current** episode because it addresses a live criticism of the model: is the present trend line staying plausible simply because refitting pulls it downward as price underperforms? We are not using this calculation to rank all historical episodes.

The decline in the fitted trend is measured against the original deficit in both:

- log space; and
- dollar price space.

Some refit drift does occur. The question is whether that drift accounts for a large or small share of the current shortfall. The results in Section 6.8 show that it accounts for only a small share.

5.8 Recovery and volatility

We test the hypothesis:

H1: declining Bitcoin volatility causes slower recovery toward the Power-Law trend.

The hypothesis is decomposed into three mechanisms.

H1a: scale

For a fixed raw residual gap, lower residual volatility necessarily produces a larger standardized deviation.

H1b: equal-standardized-gap recovery

We examine a stationary AR(1), or first-order autoregressive, residual process

$$x_t = \rho x_{t-1} + \epsilon_t.$$

At fixed ρ , dividing by the process standard deviation produces a standardized recursion whose first-passage law does not depend on the original scale parameter. The Monte Carlo experiment therefore illustrates rather than discovers an analytic scale-invariance result.

The important distinction is between a **fixed raw gap** and an **equally unusual standardized gap**. Lower volatility can make the same raw gap more extreme, but it does not by itself make recovery from the same standardized starting point slower when persistence is unchanged.

H1c: changing persistence

The remaining possibility is that persistence itself changes as Bitcoin matures.

We estimate rolling AR(1) persistence over:

- 2-year;
- 4-year; and
- 6-year

windows and compare ρ with:

- residual volatility; and
- calendar time.

Because rolling windows overlap almost completely, these correlations are treated descriptively.

5.9 Volatility time

For completed recovery episodes, calendar recovery duration is compared with cumulative absolute return accumulated during recovery.

The question is whether episodes become more homogeneous when measured in "volatility time" rather than calendar time.

5.10 Trend growth, keeping pace, and catching up

For a frozen fit,

$$P_{trend}(t) = A t^B.$$

The instantaneous log growth rate of the trend is

$$\frac{d \ln P_{trend}}{d t} = \frac{B}{t}.$$

This falls as Bitcoin ages.

We calculate two distinct return requirements.

Keeping pace

The return required to maintain the current residual is

$$\frac{P_{trend}(t+h)}{P_{trend}(t)}.$$

Catching up

The return required for current price P_t to reach the future trend is

$$\frac{P_{trend}(t+h)}{P_t}.$$

The resulting annualized rates are benchmarked against historical rolling Bitcoin CAGR distributions. Rolling-window frequencies are descriptive occupancy shares, not independent probabilities.

6. Results

6.1 Baseline reproduction and sampling robustness

The full sample yields $B=5.650938$ and $R^2=0.961281$.

Stopping the dataset at Santostasi and Perrenod's February 18, 2026 cutoff gives $B \approx 5.6912$, almost exactly reproducing their published 5.690 on 5,696 observations. Baquero and Menezes [2] obtain approximately 5.644 on a nearby CoinDesk series ending March 25, 2026. The roughly 0.05 spread across these nearby implementations and endpoints is modest relative to the exponent itself but already larger than several naive standard-error estimates, reinforcing the need to distinguish point-estimate reproducibility from precision claims.

Daily, weekly, and monthly fits produce nearly identical exponents.

These results support the claim that the empirical scaling relationship is reproducible and not sensitive to ordinary sampling frequency. They do not establish a unique structural mechanism.

6.2 The exponent is less precisely known than naive OLS suggests

The expanding-window point estimate has moved substantially over Bitcoin's history.

Selected values are:

Date	Expanding B
2016-12-31	5.6148
2018-12-31	5.8691
2020-12-31	5.7396
2021-12-31	5.8880
2022-12-31	5.8161

Date	Expanding B
2023-12-31	5.7129
2024-12-31	5.6980
2025-12-31	5.6982
2026-08-14	5.6509

The recent decline is real numerically. From the 2021 peak to August 2026, B falls by 0.2370.

However, naive OLS greatly overstates the precision of this movement.

For the current full-history estimate:

- naive OLS SE: 0.0148 **under the unrealistic assumption that residual observations are effectively independent**;
- HAC SE with 365-day lag: 0.1892; and
- moving-block bootstrap SD with 365-day blocks: 0.2401.

The HAC-based 95% interval is approximately

$$[5.28, 6.02].$$

The corresponding 365-day moving-block bootstrap 95% interval is approximately

$$[5.0466, 5.9960].$$

Prior work attaches materially narrower uncertainty to related exponent estimates. Santostasi and Perrenod [1] report $\beta = 5.69 \pm 0.05$ for the global price fit and, in a different sequential Bayesian stability exercise using local scaling estimates, a final posterior approximately $N(5.729, 0.013^2)$. Those quantities arise from different procedures than the HAC and block-bootstrap uncertainty reported here and should not be treated as directly interchangeable confidence intervals. The comparison is nevertheless informative: once serial dependence in the residuals of the global regression is allowed to affect sampling uncertainty, the slope is much less sharply pinned down than naive daily OLS suggests.

Independent work published in July 2026 reaches a closely similar conclusion by a separate route. wineLightning and Claude AI [9], in their Test 74 ("Honest error bars"), report a time-domain exponent of approximately 5.62 ± 0.21 after accounting for autocorrelation and summarize the widening as roughly 14 times naive uncertainty. Relative to our naive OLS SE of 0.0148, the 365-day HAC SE is about 12.8 times larger and the 365-day moving-block-bootstrap SD is about 16.2 times larger. The independently obtained magnitudes therefore closely corroborate one another despite differing implementations. We interpret the result here as independent corroboration rather than a novelty claim.

The 2021-2026 decline is therefore roughly:

- 1.25 HAC standard errors; and
- 0.99 365-day block-bootstrap standard deviations.

More broadly, expanding-window uncertainty falls as the sample grows:

End date	B	HAC SE
2016-12-31	5.6148	0.457
2020-12-31	5.7396	0.242
2024-12-31	5.6980	0.207

End date	B	HAC SE
2026-08-14	5.6509	0.189

The observed stabilization of B is therefore at least partly the stabilization one should expect mechanically from an expanding sample. As the history grows, each new daily observation has less leverage over the full-sample slope estimate than it did when the sample was shorter.

Interpretation

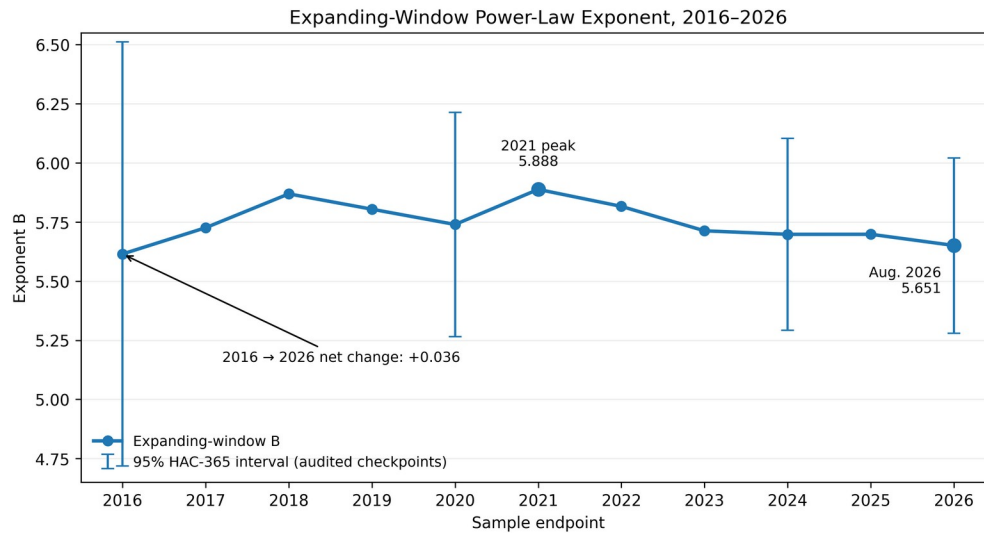
The appropriate statement is not that B has stopped changing.

It is:

The expanding estimate has become more stable, but much of that stability is consistent with increasing sample precision, and the recent decline does not by itself establish structural exponent decay.

Convergence by itself is therefore not evidence of a fixed structural exponent. An expanding estimator becomes harder to move mechanically as the sample grows, even if the underlying residual process remains highly persistent.

Figure 2. Expanding B Estimate with Robust Uncertainty



The annual estimate peaked in 2021 and has declined since, but the 2016–2026 net drift is only +0.036. Robust uncertainty remains much wider than the naive OLS error bars.

6.3 Residual persistence is extreme; stationarity evidence is mixed

The full-history residual lag-1 autocorrelation is

$$\rho_1 = 0.996985.$$

This persistence is central to almost every subsequent result.

It means:

- deviations can persist for long periods;
- adjacent daily residual observations carry little independent information because today's residual is highly informative about tomorrow's;

- standard Gaussian/independence-based probability interpretations of residual bands are inappropriate given the serial dependence;
- naive regression standard errors are too small; and
- break detectors can react to long residual cycles rather than genuine slope changes.

This serial-dependence point is separate from the expanding-sample point in Section 6.2. Persistence concerns how much new information adjacent observations contain; expanding-sample stabilization concerns how much leverage any one additional observation has on the fitted exponent.

Stationarity diagnostics are not unanimous.

ADF statistics range roughly from -2.94 to -3.61 depending on lag length. The ADF test asks, in effect, whether deviations behave like a process that remains anchored or one that can wander persistently. Several specifications reject a unit root using ordinary Dickey-Fuller critical values.

The Phillips-Perron statistic is

$$Z_t = -2.380,$$

which does not reject at conventional thresholds.

Engle-Granger-style residual statistics reject at longer lags but not at shorter ones.

We therefore characterize the evidence as:

mixed and lag-dependent.

Perrenod [3] reports rejection under all three test families on a somewhat different sample and specification. We do not interpret that difference as evidence that either analysis is necessarily "wrong." Under the present sample, construction, and lag specifications, our exact canonical series does not reproduce a clean three-test rejection.

Given the deterministic nature of $\ln t$, we do not treat the disagreement as a conventional cointegration horse race. The relevant point for this paper is simpler: the residual is extraordinarily persistent and uncertainty must be handled accordingly.

6.4 Structural-break evidence is uninformative under the matched null

Applied directly to monthly Bitcoin data, the BIC segmentation selects four breaks.

Taken at face value, that might appear to be strong evidence that the Power-Law exponent changes structurally.

The matched null tells a different story.

Under a constant true slope with realistic persistent residuals, 299 of 300 Monte Carlo simulations (99.7%) also select four breaks. The remaining simulation selects three; none select fewer.

Because the search was capped at four, these simulations do **not** tell us what a detector allowed to choose five or more breaks would have done. They do establish something narrower: even with a modest four-break cap, the detector almost always maxes out in synthetic data where the true exponent never changes.

Thus the break **count** is uninformative.

The more important audited test examines break **magnitude**.

For the real data:

- maximum adjacent slope change = 13.933; and

- slope range across segments = 16.585.

These are differences and ranges among **local segment slopes of $\ln P$ on $\ln t$** , not alternative estimates of the full-history exponent. Because an allowed segment can be as short as twelve months, local slopes can sit far from the full-history $B \approx 5.65$ and can even become negative through a drawdown. The meaningful comparison is therefore with the same local-slope statistics generated by the null simulations.

Under the constant- B null:

- the real maximum adjacent slope change is at the **54.7th percentile**; and
- the real total slope range is at the **59.0th percentile**.

These percentiles are easy to interpret: roughly half of the synthetic constant-exponent histories generate slope changes at least as large as the real ones. The real-data breaks are not sitting in an extreme tail of the null distribution; they are sitting near its middle.

Equivalently,

$$P(\text{null magnitude} \geq \text{real}) = 45.3\%.$$

The real break magnitudes are therefore ordinary under the null.

Interpretation

The result is not that we have proven there are no structural changes.

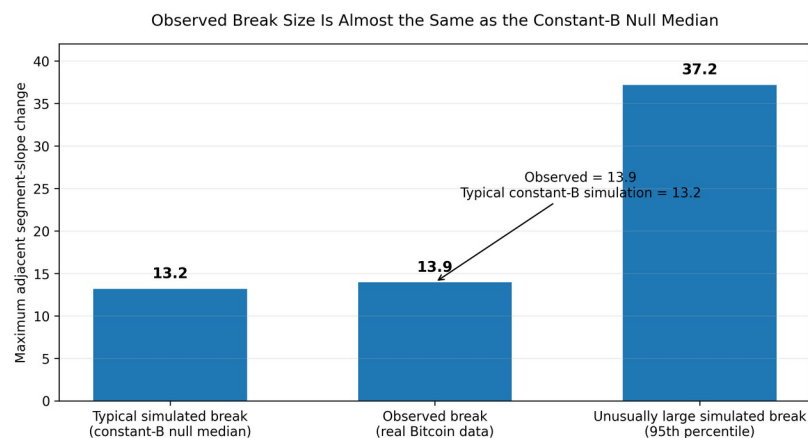
Rather:

This particular structural-break detector cannot distinguish Bitcoin's observed slope changes from false breaks generated by a constant-exponent process with persistent residuals.

Neither the number of breaks nor their size is unusual in the constant-exponent control experiment.

Baquero and Menezes [2] obtain a different result with a different statistic: their quarterly wave-stability bootstrap rejects a Power-Law-plus-AR(1) null for Bitcoin price at $p = 0.015$, while noting that the result is not Bonferroni-robust. The two tests are not direct substitutes. Their statistic asks whether stable wave components are present; ours asks whether a BIC piecewise-slope detector can distinguish the observed slope changes from those generated by a constant-exponent process with persistent residuals. The divergence is itself a reason to keep the claim narrow: structural inference is detector-dependent, and our result applies to the detector and matched null used here.

Figure 3. Structural-Break Magnitude versus Constant-B Simulations



Interpretation: the structural-break detector finds a break about as large as it typically finds even when the true exponent is held constant. A genuinely unusual break would be closer to 37 or above.

6.5 The current episode is ordinary in raw depth but unusual in modern scale

The fixed-model current below-trend episode begins on:

- November 3, 2025.

A no-lookahead real-time definition identifies approximately:

- October 28, 2025.

The dates differ because they answer slightly different questions. The fixed-model date is a retrospective classification using the fully specified episode rule, while the no-lookahead date asks when the episode would have been identified using only information available at the time. The roughly one-week difference is therefore a timing-definition effect rather than a contradiction.

Through August 14, 2026, the episode has lasted:

- 285 days under the fixed-model definition; and
- 291 days under the real-time definition.

The **deepest raw residual within the current episode** occurred on June 30, 2026:

$$r_{min} = -0.8165.$$

By August 14:

$$r = -0.7833.$$

The current episode is therefore no longer at its own maximum depth.

Historically, its raw rankings are mid-field:

- duration: #4 of 10;
- average depth: #5 of 10;
- maximum depth: #6 of 10; and
- cumulative below-trend area: #5 of 10.

Nothing about the **raw** gap is unprecedented. The current episode is still open, however. Its duration and cumulative area are right-censored and can only increase before a future trend crossing. Their final severity ranks can therefore remain unchanged or move toward #1.

The modern-volatility comparison changes the picture.

At the June 30 minimum:

- trailing four-year residual SD = 0.2989;
- local-sigma trend depth = -2.732;
- local de-meaned z = -1.907; and
- RMS trend depth = -2.107.

Among seven episodes with sufficient trailing history:

Metric	Rank by minimum
local-sigma trend depth	#1 of 7
local de-meaned z	#1 of 7
RMS trend depth	#1 of 7

The key contrast is simple: **the current episode is mid-pack in raw historical distance, but it is the deepest modern episode when the same distance is scaled by today's much smaller residual fluctuations.**

The strength of the ranking, however, differs substantially by definition.

Under the conventional locally de-meaned z-score:

- current minimum = -1.907;
- 2022 minimum = -1.847.

That is a near tie.

Under the trend-centered local-sigma measure:

- current = -2.732;
- 2022 = -1.594.

That is a much larger difference.

Why?

Because the trailing four-year residual distribution has itself shifted downward.

The trailing mean was:

- **+0.1294** at the 2022 minimum;
- **-0.2467** at the June 2026 minimum; and
- **-0.2590** by August 14, 2026.

This is not a statistical nuisance to be wished away. It is part of the phenomenon.

The two measures answer different questions.

The de-meaned z-score asks:

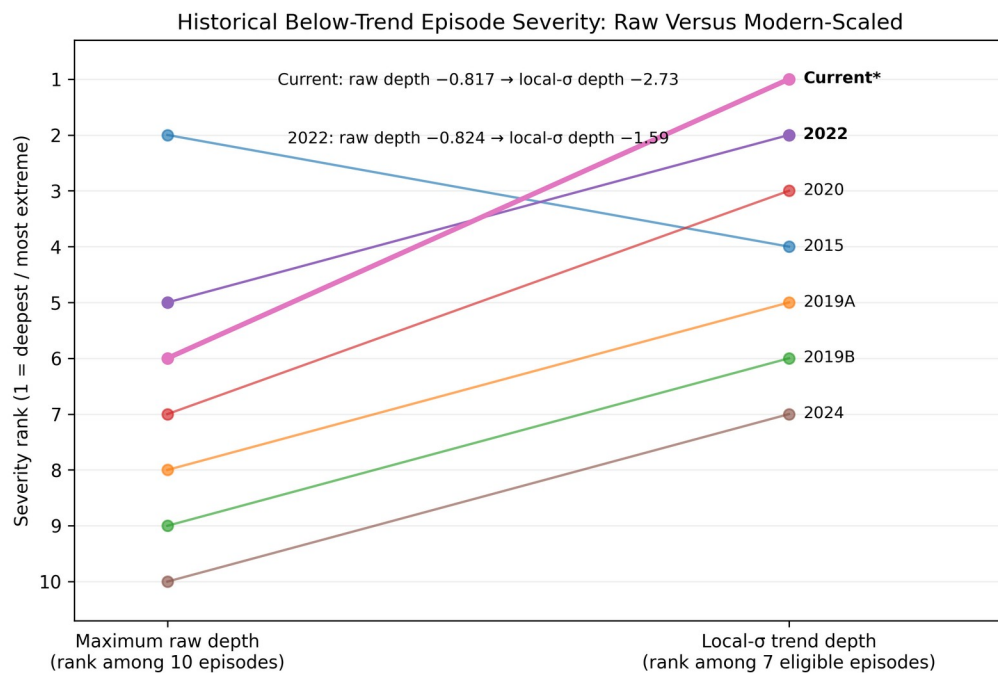
How unusual is this observation relative to where residuals have recently been centered?

The local-sigma trend depth asks:

How far below the Power-Law model's zero line is price relative to the amount residuals typically fluctuate in the modern market?

For testing the Power-Law trend itself, both are useful.

Figure 4. Historical Episode Severity: Raw versus Local-Standardized. Note: raw-depth ranks use all 10 episodes; local-sigma ranks use the 7 episodes with sufficient trailing history. Connecting lines identify the same episodes across different eligible sets, not a common-sample change in rank.



*Current episode is open/right-censored. Local-σ trend depth is raw residual divided by the strictly prior trailing-4y residual SD; it is not a de-measured z-score.

6.6 The 2022 and 2026 raw gaps are essentially the same

The current episode's raw maximum depth is

−0.8165.

The 2022 minimum was

−0.8243.

The current gap is marginally shallower.

If the current 2026 minimum is evaluated using the 2022 residual-volatility denominator, its trend depth becomes approximately

−1.579,

compared with

−1.594

for 2022 itself.

Therefore the smaller modern denominator accounts for slightly **more than the entire difference** in trend-centered local-sigma depth: the 2026 raw gap is actually marginally shallower than the 2022 gap, so the numerator works slightly against, rather than in favor of, the larger standardized reading.

An earlier internal comparison mixed different episode dates. The audited comparison above uses the matched episode minima and supersedes that preliminary result.

The correct interpretation is:

2022 and 2026 reached nearly identical raw distances below the trend. The current episode appears much more severe under a trend-centered local-volatility measure because residual fluctuations have become much smaller.

6.7 The current episode deteriorated into June, but is not currently at its low

Dividing the episode into three equal **time** segments, the successive thirds have mean raw residuals of approximately

$$-0.212, -0.531, -0.694.$$

That progression documents substantial deterioration across the episode.

However, the maximum depth was reached June 30, 45 elapsed days before the end of the sample; including both endpoints gives 46 daily observations in the post-low window.

Since that low:

- price rose approximately 7.6%;
- the residual improved by about 0.033 log units; and
- the 46-observation post-low slope is approximately flat.

Medium-horizon residual slopes remain negative:

- 90-day: approximately -0.888 log/year;
- 180-day: approximately -0.592;
- 365-day: approximately -1.017; and
- episode-wide: approximately -0.883.

These horizons describe different parts of the same path. Most of the deterioration occurred before the June 30 low; the period since then has shown limited net improvement rather than a fresh collapse or a sustained recovery.

The correct description is therefore:

The episode deteriorated steadily into the June 30 low and has not yet established a sustained recovery. It is not, however, at its worst reading as of August 14, and the post-low period is approximately flat rather than still deteriorating at the earlier pace.

6.8 Refitting explains only a small part of the current deficit

One concern with a continuously refitted Power Law is that the model might appear stable simply because underperformance gradually pulls the fitted trend downward.

We test this directly over the current episode.

At episode start:

$$B=5.703364.$$

At the end of the sample:

$$B=5.650938.$$

The corresponding current-date trend values are approximately:

- episode-start frozen fit: **\$147.4k**; and
- current refit: **\$137.9k**.

The refit lowers the trend by approximately:

- 6.45%.

Measured against the original deficit, refitting absorbs:

- **7.84%** in log space; and
- **11.26%** in price space.

In dollar terms, approximately \$9.5k of the \$84.4k **current-date gap measured against the frozen episode-start fit** is absorbed by the refit.

Thus:

During the current episode, the overwhelming majority of the deficit remains actual price underperformance relative to the episode-start model rather than an artifact of the model moving downward.

There is some refit drift. The important result is that the drift is small relative to the shortfall it is sometimes accused of explaining.

This statement is deliberately limited to the current episode.

6.9 Declining volatility does not provide a supported explanation for slower recovery

The original recovery hypothesis was:

Lower Bitcoin volatility makes return to the Power-Law trend systematically slower.

The hypothesis sounds plausible but combines several distinct ideas.

Scale effect

From a fixed raw gap, smaller residual volatility mechanically creates a larger standardized deviation.

This is true by definition.

Equal-standardized-gap effect

Suppose residuals follow

$$x_t = \rho x_{t-1} + \epsilon_t.$$

Let S_x be the stationary standard deviation and define

$$u_t = \frac{x_t}{s_x}.$$

Then u_t follows the same recursion with the same ρ and standardized innovation variance $1 - \rho^2$, regardless of the original value of s_x . The scale parameter has been divided out: two stationary AR(1) processes with different volatility scales but the same persistence generate the same standardized dynamics.

At fixed ρ , first-passage behavior from a starting point such as $-2s_x$ therefore does not depend on the original volatility scale.

Monte Carlo simulations illustrate the analytic result.

At **equal raw depth**, lower-volatility regimes produce longer median crossing times. That happens because the same raw gap represents a more extreme standardized displacement when the usual fluctuations are smaller.

At **equal standardized depth**, simulated median recovery times are approximately

$$398, 393, 392 \text{ days}$$

across the three volatility regimes.

The second comparison is the relevant test of the proposed scale mechanism within this conditional AR(1) framework. Once the starting deviation is equally unusual in each regime, changing the volatility scale alone does not slow recovery.

The simulation should therefore be understood as a numerical illustration of scale invariance, not empirical evidence about Bitcoin.

Persistence effect

A genuine maturation mechanism could still exist if ρ itself changed systematically as volatility declined.

Rolling estimates do not show a clean pattern.

Using four-year windows, era-average ρ values remain approximately

$$0.9961 \text{ to } 0.9974.$$

Correlations between residual SD and ρ change sign depending on window length:

- 2-year: positive;
- 4-year: weakly negative; and
- 6-year: more negative.

Within individual eras, signs also change. Estimated half-lives are non-monotonic.

Across the reported four-year era-average persistence estimates, the corresponding conditional AR(1) half-lives range from roughly 177 to 266 days. Thus, small-looking changes in ρ near unity can correspond to materially different mean-reversion times.

The longer four- and six-year windows lean in the H1-consistent direction: lower residual volatility is associated with higher estimated persistence, while the two-year window points the other way. Because the rolling windows overlap heavily, the sign and magnitude are not stable across reasonable window choices, and within-era signs also change, we do not treat the longer-window correlations as establishing a persistence channel.

Historical completed episodes likewise provide little identifying power, but what signal they contain does not support H1. Across the nine completed episodes, trailing residual volatility versus days to recovery has Spearman correlation of approximately **+0.14**, opposite the negative sign H1 would predict. A depth-controlled regression also produces a positive volatility coefficient, but with so few episodes the estimate is unstable; leave-one-out relationships change sign depending on which episode is omitted. Depth is more stable as an ordering variable than volatility.

The AR(1) exercise is conditional on a stationary process with $|\rho| < 1$. Because Section 6.3 finds mixed unit-root evidence, we do not claim that Bitcoin residuals are empirically established to follow a stationary AR(1). The exercise isolates one proposed mechanism: **within the specified stationary AR(1) class**, changing scale alone cannot change recovery from an equally standardized starting point.

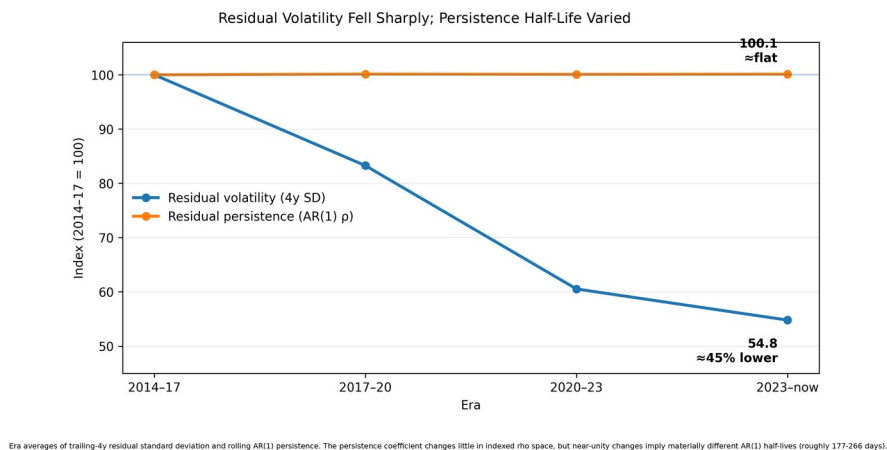
Final verdict

The hypothesis that declining volatility slows Power-Law recovery is not supported.

More precisely:

- the scale component is true but tautological;
- slower recovery from equal standardized depth is false under fixed-persistence AR(1); and
- a persistence-driven maturation effect is not established and cannot be cleanly excluded.

Figure 5. Residual Volatility Fell Sharply while Four-Year Persistence Remained High but Implied Half-Lives Varied



6.10 Volatility time offers, at most, a modest secondary effect

Recovery episodes were also measured using cumulative absolute return rather than calendar time.

The coefficient of variation falls from:

- 0.884 in calendar time; to
- 0.740 in volatility time.

This suggests that part of the calendar-time variation may reflect volatility arriving at different speeds.

With only nine completed episodes, we do not regard this reduction as reliably distinguishable from ordinary sampling variation.

We therefore treat volatility time as a secondary descriptive result rather than support for the main recovery hypothesis.

6.11 The trend's required growth rate is declining

For

$$P_{trend}(t) = At^B,$$

the instantaneous log growth requirement is

$$\frac{B}{t}.$$

At the end of the sample:

- annualized log growth = approximately 0.3209; and
- instantaneous arithmetic equivalent = approximately **37.84%**.

Exact frozen-fit CAGRs are:

Horizon	Trend CAGR
1 year	36.63%
2 years	35.52%
3 years	34.49%
5 years	32.64%
10 years	28.93%

The required rate falls as Bitcoin ages.

From 2012 to the present:

- annualized trend log-growth requirement falls approximately **77.3%**; and
- annualized return-volatility SD falls approximately **67.6%**.

A movement-scale ratio,

$$\frac{\text{return SD}}{\text{trend log-growth requirement}},$$

therefore rises from approximately

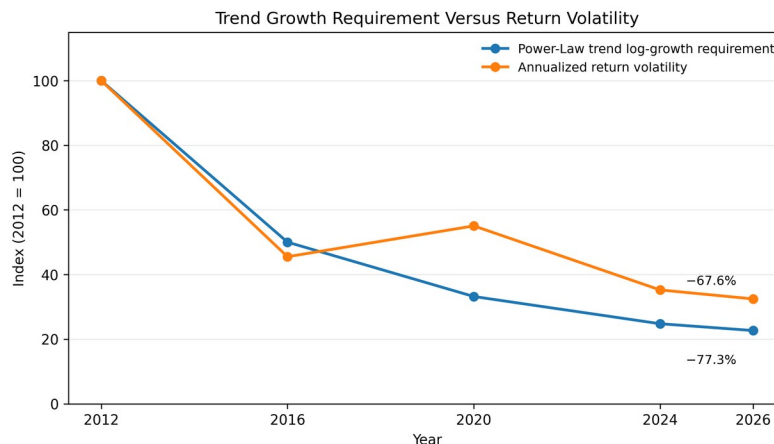
$$0.962 \rightarrow 1.377.$$

This comparison should not be misunderstood.

Volatility is dispersion, not expected return. It tells us how widely returns move around, not how fast their average path moves upward. An asset can be extremely volatile - very bumpy - and still fail to generate enough average upward drift to meet a growth target.

The result establishes only this: the trend's required log-growth rate has fallen faster than the size of Bitcoin's typical return fluctuations. The distribution of yearly price movement is therefore not mechanically narrower relative to the trend hurdle than it was earlier in the sample. That says nothing about whether the **average or realized** return is high enough to clear the hurdle. Section 8 examines that growth question separately.

Figure 6. Trend Growth Requirement versus Return Volatility



Selected audited checkpoints. Intermediate trend values use the reported rounded arithmetic CAGR checkpoints converted to log growth; 2012 and 2026 endpoints use the exact audited log-growth values.

7. Keeping Pace Versus Catching Up

Perhaps the clearest result of the study is the distinction between two superficially similar questions.

Question 1

How fast must Bitcoin grow to remain the same distance below the Power-Law trend?

Question 2

How fast must Bitcoin grow to erase its existing deficit and reach the trend?

They are very different problems.

7.1 Keeping pace

The exact next-one-year trend CAGR is

36.63 %.

Across Bitcoin's full historical rolling one-year return distribution, that hurdle is near the:

- **35th percentile.**

Approximately:

- 65% of all historical rolling one-year windows exceeded it;
- 60% of windows beginning on or after January 1, 2016 exceeded it; and
- 55% of windows beginning on or after January 1, 2020 exceeded it.

Here and in the rest of Sections 7-8, era labels such as "post-2020" are assigned by **window start date**. Section 7 applies **today's fixed hurdle** to historical realized returns; Section 8 asks a different question by comparing each historical window with the hurdle that prevailed at that window's own start.

These percentages are not forward-looking probabilities. Rolling windows overlap strongly.

They simply show that a one-year return on the order required to **keep pace** has not been unusual historically.

7.2 Catching up

Closing the existing gap requires much larger returns.

Approximate annualized requirements are:

Horizon	CAGR required to reach trend
1 year	199.0%
2 years	100.5%
3 years	74.6%
4 years	62.4%
5 years	55.1%

To put those requirements in historical context, we ask what share of post-2020 rolling windows actually produced a realized CAGR **at least as high as the catch-up rate shown above**.

The descriptive shares are approximately:

- 1-year catch-up rate met or exceeded: **17%**;
- 2-year catch-up rate met or exceeded: **18%**; and
- 3-year catch-up rate met or exceeded: **about 9.0%**.

These horizon-specific occupancy shares need not be monotonic: both the realized-return distribution and the eligible start-date sample change with horizon.

The take-home is simple: **keeping pace has been historically common; catching up from the present deficit has been historically much rarer**.

The eligible post-2020 start-date span covers only about 1.21 non-overlapping three-year window lengths, so the three-year number should be treated with considerable caution. The post-2020 five-year comparison is suppressed because its eligible start dates represent only about **0.32 independent-window equivalents**, too little history to make the percentage useful.

These are historical occupancy shares, not forward probabilities.

7.3 Interpretation

The current difficulty is therefore not adequately described as:

"Can Bitcoin still grow at roughly 35-40%?"

The more demanding question is:

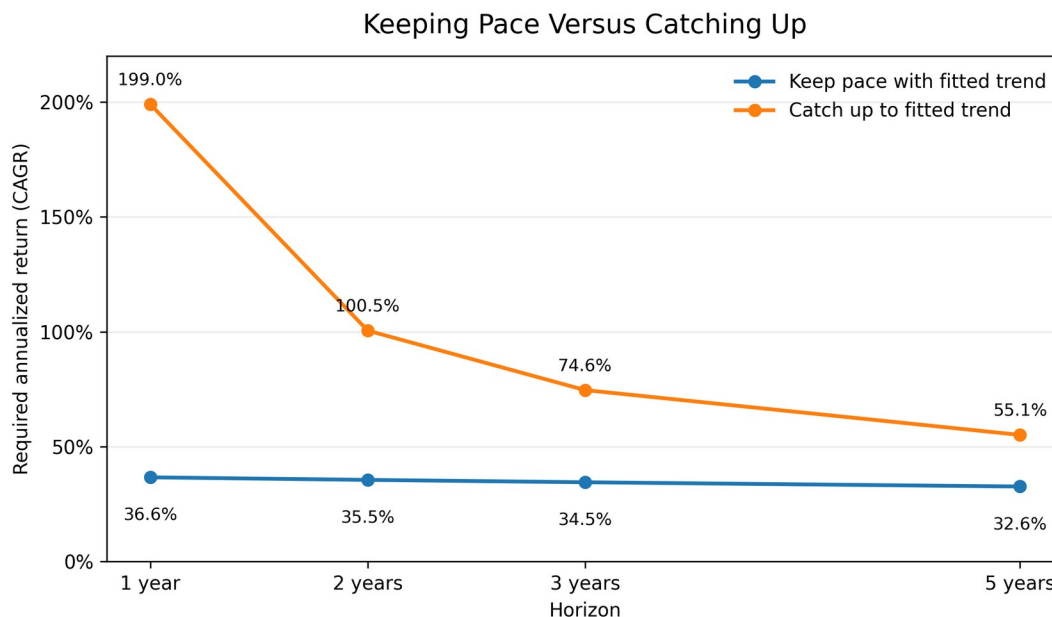
"Can Bitcoin grow fast enough, for long enough, to eliminate a roughly 0.78-log accumulated deficit while the trend itself continues rising?"

That distinction helps reconcile two observations that otherwise appear contradictory.

The trend's required growth rate is gradually becoming easier.

Yet return to trend can simultaneously become more difficult if a large deficit has already accumulated.

Figure 7. Keeping Pace versus Catching Up



Frozen at August 14, 2026. "Keep pace" preserves the current residual; "catch up" reaches the fitted Power-Law trajectory by the stated horizon.

8. A Separate Open Question: Realized Growth Versus the Contemporaneous Hurdle

The volatility investigation produced another observation that should not be conflated with the volatility-recovery hypothesis.

For rolling windows whose **start dates are on or after January 1, 2020**, we compare realized CAGR with the Power-Law growth hurdle that was actually in force at each window's own start:

Horizon	Median realized CAGR	Median contemporaneous hurdle	Share beating own hurdle
1 year	~56%	~48%	~52%
2 years	~42%	~49%	~46%
3 years	~35%	~49%	~26%

Thus the one-year comparison remains favorable on median, while the two- and especially three-year comparisons are less so. This is intentionally different from Section 7.1, where today's fixed 36.63% keep-pace hurdle is applied to the historical one-year return distribution.

That 26% figure does **not** mean that Bitcoin has been "above the Power Law only 26% of the time since 2020." It is specifically a comparison of realized three-year growth rates with the growth hurdle that existed at each window's start.

This is potentially important, but the independent sample is extremely small.

A similar weak-growth period also appears in an earlier 2018-2022 comparison and subsequently resolved.

The correct conclusion is therefore not that a structural maturation squeeze has been identified.

It is:

Recent Bitcoin has at times failed to generate enough sustained multi-year realized growth to match the contemporaneous Power-Law hurdle. The available independent history is insufficient to determine whether this is cycle-specific noise or the beginning of something structural.

We leave this observation open rather than adding another research branch to the present study. It is a natural candidate for future work as additional independent multi-year data accumulate, and other researchers may wish to test it with different methods.

9. Discussion

9.1 What survives the stress tests

Several properties of the empirical relationship remain intact.

First, the regression itself is highly reproducible. Our estimate at the Santostasi-Perrenod cutoff is essentially identical to theirs.

Second, the exponent is robust to ordinary changes in sampling frequency.

Third, apparent structural breaks lose evidentiary force when the detector is applied to an appropriate persistent-residual null. Both the number and magnitude of detected breaks are ordinary under that null.

Fourth, the current model is not simply eliminating its own error through rapid refitting. During the current episode, approximately 89% of the original dollar deficit and 92% of the original log deficit remain after refitting.

Fifth, the trend's required percentage growth falls with time rather than rising indefinitely.

None of these findings proves that the model will continue to describe Bitcoin indefinitely.

What survives is the empirical description under the tests completed here - not proof of a law of nature.

The tests do, however, weaken or rule out several narrower failure explanations within the scopes tested.

9.2 What does not survive

The claim that the expanding Bitcoin exponent has converged to a sharply known constant does not survive robust uncertainty analysis.

The current estimate is far less precise than naive OLS suggests.

Similarly, our exact residual series does not yield a clean, unanimous stationary-residual result across ADF, Phillips-Perron, and residual-based tests.

The claim that lower volatility itself explains slower recovery also does not survive decomposition of the mechanism.

And the idea that the current raw deficit is historically unprecedented is false.

The latter distinction matters: the present episode is unusual relative to the **modern volatility regime**, not because its raw log gap exceeds all historical precedents.

9.3 What is genuinely unusual about the current episode

The current episode is unusual not because Bitcoin has traveled farther below the trend in absolute log distance than it did in 2022.

It has not.

It is unusual because:

1. the same raw gap has occurred in a much lower residual-volatility environment;
2. the recent residual distribution itself is centered materially below the trend;
3. the episode became progressively weaker through June 2026; and
4. the current price must both keep pace with a rising trend and close a large accumulated deficit.

These points are distinct.

The first creates the extreme local-sigma trend-depth reading.

The second explains why that reading looks much less extreme after local de-meaning.

The third documents the path into the current low.

The fourth explains why a declining future trend hurdle does not automatically make near-term recovery easy.

9.4 Current episode as a live test

Prior work has already described the 2025-2026 period as a live test of the Bitcoin Power Law. We agree with the framing but treat it quantitatively rather than rhetorically.

At the August 14, 2026 freeze:

- price is approximately \$63.0k;
- fitted trend is approximately \$137.9k - the model-implied Power-Law trend level;
- raw residual is -0.7833;
- trend-centered local-sigma depth is -2.521; and
- locally de-meaned z is -1.687.

The largest historical cumulative below-trend episode had **area**, not point depth, of approximately

537.6 log-days.

The current episode has accumulated approximately **137 log-days** through the cutoff, well below that completed historical maximum in cumulative area. This does not contradict its unusually large modern-standardized depth: one metric measures how deep the episode gets at a point; the other measures how much below-trend deficit accumulates over time.

A duration beyond approximately March 22, 2028 without a trend crossing would exceed the longest prior episode under the current episode definition.

These thresholds are not forecasts and should not be treated as magical falsification points. They are pre-specified empirical markers that would make the current episode increasingly difficult to describe as ordinary historical persistence.

9.5 Falsifiability matters more than narrative

The Power Law is easiest to misuse when every new outcome is made consistent with it after the fact.

That is why the current episode is valuable analytically.

A useful empirical framework should state **in advance what future observations would reduce confidence in it**. Falsifiability does not mean hoping the model fails; it means refusing to move the goalposts after unfavorable evidence arrives.

The monitoring markers fixed before publication include:

- no trend crossing beyond roughly **March 22, 2028**, corresponding to a duration beyond the prior ~870-day maximum under this episode rule;
- cumulative below-trend area above approximately **537.6 log-days**;
- a trailing four-year residual mean below approximately **-0.40 and still falling**;
- continued residual deterioration combined with a materially larger downward movement in B , beyond the scale already observed, while residuals remain negative; and
- persistent multi-year realized growth below the contemporaneous hurdle across genuinely new independent windows.

Not every future warning can be reduced to a single timeless cutoff: robust uncertainty around B , for example, will itself evolve as new data arrive. The point is to specify the form of adverse evidence in advance rather than invent a new rescue narrative after it occurs.

The model should not be "rescued" by inventing explanations only after those events occur.

One further question lies outside the present study: whether observed residual deficits can be related to economically plausible capital flows. Future work could examine the scale of net allocation or systematic rebalancing demand associated with closing a given Power-Law residual, conditional on market capitalization, liquidity, and the size of the addressable capital pool.

10. Limitations

10.1 Time-origin dependence

The fitted exponent depends materially on the time origin. Baquero and Menezes [2] report nearly a threefold change in the fitted exponent across an additive time-shift sweep, although their information criterion is minimized at the unshifted genesis convention.

We use Bitcoin genesis because it defines the model being studied, not because the exponent is mathematically invariant to alternatives.

Any structural interpretation of the numerical value of B must acknowledge this dependence.

10.2 Persistent residuals

Residual autocorrelation near 0.997 creates severe inferential complications.

HAC estimates and block bootstraps help, but no single bandwidth or block length is uniquely correct.

The uncertainty estimates reported here are therefore better viewed as a robust range than a precise confidence statement.

10.3 Stationarity ambiguity

Our residual diagnostics are mixed.

The Phillips-Perron result does not reject a unit root, and residual-based rejection depends on lag specification.

This limits strong claims that deviations are governed by a stable stationary mean-reverting process.

It also qualifies the AR(1) mechanism illustration in Section 6.9. That exercise assumes a stationary AR(1) with $|\rho| < 1$ so that a stable process variance exists and standardized recovery can be compared across scales. We use it to isolate the volatility-scale channel, not to assert that Bitcoin residuals have been proven to follow that process. If the true residual dynamics are unit-root-like or otherwise nonstationary, the mean-reversion interpretation itself becomes weaker; the AR(1) result remains a conditional statement about the mechanism tested.

10.4 Small number of completed episodes

Only nine completed recovery episodes are available, with still fewer possessing full local-volatility history.

Episode-level correlation statistics are exploratory. Put simply, nine historical episodes are not enough to estimate a precise population relationship between volatility, depth, and recovery time. These correlations are descriptive rather than precise population inferences with conventional statistical significance.

10.5 Overlapping return windows

Rolling one-, two-, three-, and five-year return windows overlap substantially **within each horizon**.

For example, two consecutive three-year windows that start one day apart share almost all of their observations. They are therefore not independent three-year experiments simply because a computer can calculate thousands of them.

Percentages of windows exceeding a hurdle measure historical occupancy, not independent success probabilities.

This is especially important for post-2020 multi-year comparisons.

10.6 Structural-break detector limits

The segmentation algorithm was capped at four breaks and selected that boundary.

Our conclusion is therefore not that the optimal number of breaks equals four.

Rather, the detector itself is shown to produce similar behavior under the constant-slope null and is consequently uninformative for the structural question in its current form.

Higher break caps were outside the scope of this study.

10.7 Simulation scope

The AR(1) simulations are mechanism illustrations.

They do not reproduce Bitcoin's:

- heavy tails;
- asymmetric returns;

- changing persistence;
- cycle structure;
- changing drift; or
- market-regime shifts.

Their useful result is analytic: scale alone cannot change standardized first-passage behavior when persistence is fixed.

10.8 No independent forecasting study

This paper studies empirical stability, residual behavior, and recovery conditions.

It does not conduct a new out-of-sample forecasting tournament.

Baquero and Menezes [2] already provide a more direct analysis of that question.

10.9 No claim of causal mechanism

The analysis is primarily price-based.

We do not estimate the network-adoption and generalized Metcalfe components required to establish the causal or mechanistic interpretation proposed in other work.

In other words, reproducing and stress-testing the price-age relationship does not tell us whether network effects are the causal reason that relationship exists.

10.10 Local-standardization window

The four-year trailing window is the local-standardization convention used in this study. The headline raw-versus-local comparison is therefore conditional on that window choice. We did not run a new sensitivity sweep of episode rankings over alternative two-, four-, or six-year local-volatility denominators after the research freeze; that is a natural subject for future work rather than a result implied here.

10.11 Residual-model alternatives

The break-null and recovery simulations treat residual structure as persistent stochastic variation, using AR-type dependence as the mechanism under study. Other work models Bitcoin's deviations with deterministic or quasi-deterministic log-periodic components [7,8]. We do not adjudicate between those residual specifications in this paper. Our conclusions about break-detector behavior and AR(1) scale invariance are conditional on the residual models actually tested.

10.12 Episode-definition sensitivity

The episode inventory and all episode-derived rankings, recovery comparisons, and forward monitoring markers are conditional on the stated 60-day minimum-duration rule and the strict-crossing/bridge conventions in Section 5.5. The 3-day and 7-day bridge tolerances were used as robustness checks for short interruptions around zero, but this study did not sweep alternative minimum-duration thresholds such as 30 or 90 days. Accordingly, quantities such as the episode count, historical severity ranks, longest-prior-episode duration, and cumulative-area benchmark should be read as conditional on the canonical episode definition rather than invariant features of the data.

11. Conclusion

Bitcoin's historical price remains descriptively close to a power-law function of its age under the conventional genesis-date time origin: the log-log regression has $R^2=0.961$, while one full-history residual standard deviation still corresponds to roughly a factor-of-two deviation in price.

That descriptive statement survives.

Several stronger statements do not.

The fitted exponent is less precisely known than naive OLS suggests. Its apparent expanding-window stabilization is partly a mechanical consequence of accumulating data. The **post-2021 decline in the expanding estimate of B** is real numerically, but it does not on its own establish structural decay. The longer comparison is also less dramatic: the 2026 estimate remains slightly above the 2016 estimate, reinforcing that a multi-year downward run should not automatically be read as permanent exponent decay.

Statistical break detection likewise requires caution. A detector that appears to find four historical breaks also finds four breaks almost universally in constant-exponent simulations with persistent residuals, and the magnitudes of the real breaks are ordinary under that null.

The current 2025-2026 below-trend episode is not unprecedented in raw distance. Its maximum raw residual is effectively the same as the 2022 minimum.

What has changed is the environment around that gap.

Residual volatility is substantially lower, and the recent residual distribution itself is centered materially below zero. The current episode is therefore extreme when measured as distance from the Power-Law trend relative to modern residual fluctuation size, but only marginally more extreme than 2022 under a conventional locally de-meaned z-score.

This distinction matters.

A trend-centered standardized measure asks whether price is far from the model.

A de-meaned measure asks whether price is far from recent experience.

The current market is more unusual under the first question than the second because recent experience has itself shifted below the long-run trend.

The hypothesis that declining volatility causes slower recovery is not supported. Lower variance makes a fixed raw gap larger in standardized units, but it cannot alter equal-standardized-gap recovery under fixed-persistence AR(1) dynamics. In plainer terms: if two episodes start equally far from trend **relative to the volatility of their own eras**, simply lowering the volatility scale does not make the lower-volatility episode recover more slowly. Persistence itself does not exhibit a sufficiently stable maturation pattern in our data to restore the hypothesis.

The trend's own growth requirement is also declining. Bitcoin does not need an ever-increasing percentage growth rate to remain on a t^B trajectory.

But that fact should not be mistaken for an easy recovery path.

At the current deficit, keeping pace requires roughly 36.6% over the next year. Reaching the trend requires approximately 199% over one year, 100.5% annualized over two years, or 74.6% annualized over three.

The accumulated deficit - how far price currently sits below the model-implied trend level - not simply the slope of the trend, is the binding constraint.

The resulting conclusion is deliberately limited:

Bitcoin's historical price continues to exhibit a strong descriptive Power-Law relationship with age under the specified time-origin convention. The completed robustness tests do not provide compelling evidence of structural failure, but they also do not establish an immutable law. The current below-trend episode remains a meaningful live test: raw underperformance is historically familiar, modern trend-centered depth is unusually large, the recent residual center is persistently negative, and whether future realized growth is sufficient to close the deficit remains unresolved.

That unresolved part is not a defect in the analysis.

It is the question future data will answer.

Code and data availability. Code and the frozen analysis dataset used to reproduce the reported results will be released publicly with the final publication version.

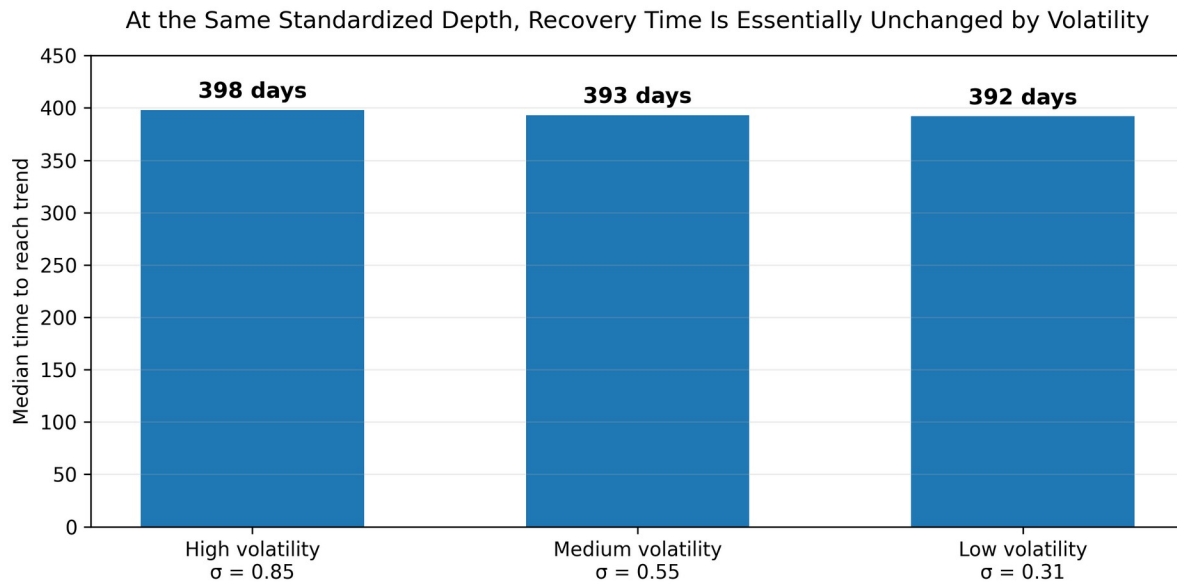
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Appendix A. Canonical Numerical Snapshot and Conditional AR(1) Illustration

Quantity	Audited value
observations	5,873
sample	2010-07-17 to 2026-08-14
A	4.156832×10^{-17}
B	5.650938
R^2	0.961281
full-history residual SD	0.697088
residual lag-1 autocorrelation	0.996985
HAC SE of B , lag 365	0.18916
365-day block-bootstrap SD of B	0.2401
365-day block-bootstrap 95% interval	[5.0466, 5.9960]
current completed close	\$62,996.40
current trend	~\$137.9k
current raw residual	-0.7833
current local-sigma trend depth	-2.521
current local de-meaned z	-1.687
current RMS trend depth	-1.936
current trailing-4y residual mean	-0.2590
current trailing-4y residual SD	0.3107
episode maximum raw depth	-0.8165
maximum-depth date	2026-06-30
episode-minimum local-sigma trend depth	-2.732
episode-minimum de-meaned z	-1.907
current cumulative below-trend area	~137 log-days
largest prior episode area	537.6033 log-days
arithmetic equivalent of instantaneous annualized log-growth rate	37.84%
exact one-year trend CAGR	36.63%
one-year catch-up CAGR	199.0%
two-year catch-up CAGR	100.5%
three-year catch-up CAGR	74.6%
refit absorption, log space	7.84%
refit absorption, price space	11.26%

Appendix Figure A1. Equal-Standardized-Gap Recovery Is Scale-Invariant under Fixed-Persistence AR(1)



Range: 6 days (~1.5%). The small spread is simulation variation, not a modeled volatility effect.
Conditional AR(1) illustration with fixed persistence and the same starting depth in standard-deviation units.

Appendix B. Claims Deliberately Not Made

This paper does not claim that:

- the numerical exponent is invariant to time-origin choice;
- a high R^2 proves a structural law;
- residuals cleanly reject a unit root under every specification;
- the residual process has been proven to be a stationary AR(1);
- the break analysis proves that structural change is impossible;
- the local-sigma trend-depth statistic is a Gaussian z-score;
- the four-year local-standardization ranking is invariant to other trailing-window choices;
- declining volatility has been proven incapable of affecting recovery through every possible mechanism;
- historical rolling-window occupancy rates are forward probabilities;
- recovery from prior below-trend episodes implies recovery from the current one;
- the current-episode refit decomposition generalizes to all historical periods;
- Santostasi and Perrenod's mechanistic derivation has been validated by reproducing their regression; or
- the Power Law guarantees any future Bitcoin price.